# Andrew C. Stewart, AEP

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## PROFESSIONAL SUMMARY

Analytical, research-driven associate ergonomics professional with a Master of Science in Industrial and Systems Engineering, concentrating in human factors & ergonomics. Experienced in human subjects research and ergonomic workplace assessment, including a thesis examining the effects of augmented reality on postural balance during simulated ladder-based workplace tasks; findings have been accepted for publication by the leading AR/VR journal.

## EDUCATION

**Master of Science — Industrial & Systems Engineering, Human Factors & Ergonomics Concentration**
*Virginia Polytechnic Institute and State University, Blacksburg, VA,* August 2024 – May 2026, GPA: 3.76

***Thesis — Effects of Performing Augmented Reality Tasks on Postural Sway While on a Ladder***
**Accepted for The 25th IEEE International Symposium on Mixed and Augmented Reality (11% acceptance rate)**

- Examined AR's effects on postural balance for construction workers and inspectors performing tasks on ladders
- Designed a study where participants completed four simulated ladder-based work tasks with AR assistance; data collected and analyzed in MATLAB

**Bachelor of Science — Industrial & Systems Engineering, Human-Computer Interaction Minor**
*Virginia Polytechnic Institute and State University, Blacksburg, VA,* August 2020 – May 2024, Overall GPA: 3.32, In-Major GPA: 3.56

## PROJECT/RESEARCH EXPERIENCE

**Quality of Life Plus Student Engineer, Interdisciplinary Design Course @ VT** *Fall 2023 – Spring 2024*

- Collaborated on a senior design project with Quality of Life Plus, a nonprofit serving veterans and first-responders with injuries or disabilities
- Designed and built a lower limb exoskeleton enabling the client to walk comfortably and maintain active lifestyle
- Managed the team's budget and oversaw purchasing of components
- Oversaw safety considerations throughout exoskeleton development

**Research Assistant, Madigan Biomechanics Group** *Spring 2024*

- Investigated slip recovery training as a tool for workplace injury prevention
- Conducted tracking analysis on experiment recordings and managed subsequent database entry
- Supported participants through human-subjects testing procedures

**Research Assistant, Mind Music Machine Lab** *Spring 2023*

- Designed novel XR interactions for a 3D planetary model exhibit (“Large Scale Object Manipulation in XR”)
- Surveyed literature on virtual reality and interactive arts to inform design decisions
- Designed and tested new 3D VR interaction methods through a user experiment
- Analyzed data using statistical tools and presented findings

## SKILLS

- **Human factors & ergonomics:** workplace assessment, biomechanical analysis, injury prevention research
- **Human subjects experimental research:** IRB-compliant protocols, behavioral and social research methods
- **Human-centered design & HCI:** UI/UX principles, usability evaluation, AR/XR interaction design
- **Data collection, management, & analysis:** Excel data management; MATLAB data collection and analysis
- **Communications:** oral presentations, live experimental demos, corresponding with stakeholders

## AFFILIATIONS

Human Factors and Ergonomics Society (2024–Present); Cognitive Engineering for Novel Technologies (COGENT) Lab (2025–2026); Madigan Biomechanics Group (2024–2026); Mind Music Machine Lab (2023)

## LEADERSHIP

Senior Staff Assistant, The Marching Virginians (2025); Vice President of Membership, Kappa Kappa Psi (2023); Section Leader, The Marching Virginians (2023–2025); Public Relations Officer, The Marching Virginians (2022); Rank Captain, The Marching Virginians (2021–2022)

## QUALIFICATIONS

Associate Ergonomics Professional, Board of Certification in Professional Ergonomics (2026);Order of the Engineer (2024); CITI Social and Behavioral Research Certified (2022, 2025); CITI Basic Responsible Conduct of Research Certified (2024)